

Osaid Khan

+1 (945) 304-5781 | khanosaid726@gmail.com | www.linkedin.com/in/-osaid-khan/ | github.com/ozzyozbourne

Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **Software Development Engineer** role with **Amazon Web Services (AWS)**. I bring over three years of professional experience building distributed systems, complex cloud-native backends, and production AI agents, along with a strong academic foundation as a Master of Science in Computer Science graduate (GPA 4.0/4.0) from Pace University, NY.

Through my experience across two engineering roles, I have built and delivered production-grade distributed and multi-tiered systems that directly reflect the depth of designing and developing large-scale, multi-threaded, distributed software applications this position demands. As a Software Engineer at Sperse, I designed a multi-agent CRM platform in Python, LangGraph, and FastAPI that serves 10,000+ users by routing requests across 40+ specialized sub-agents via a distributed systems task-routing architecture with tenant-aware access control — a complex software system engineered for horizontal scaling and reliability. Alongside it, I engineered a MCP server using FastMCP exposing 160+ LLM-callable backend actions over a multi-tiered service, with built-in monitoring and Pydantic validation to ensure secure, reliable execution of cloud computing workflows across multiple customers. To keep these systems production-ready, I built GitHub Actions CI/CD pipelines integrating OpenTelemetry trace-based eval suites that catch latency regressions and safety issues before every deploy. Prior to Sperse, at Qualitest, I designed a cloud-agnostic Storage Service in Java and Spring Boot with PostgreSQL and Redis handling file operations across AWS S3, Google Cloud Storage, and MinIO via expiring signed URLs — and orchestrated long-running jobs with AWS Step Functions to eliminate Lambda timeout failures across 100K+ weekly executions, applying architecture and design patterns focused squarely on fault tolerance and scaling of complex software systems. Together, these roles have prepared me to design, develop, and maintain efficient, reusable, and reliable code at the scale AWS demands, and to contribute meaningfully to cross-functional feature work, architecture reviews, and mentoring within an engineering team.

My academic and personal project experience further demonstrates the operational excellence and passion for invention that AWS Leadership Principles require. As part of my graduate work, I built the Dark Factory coding-agent orchestrator in Elixir and TypeScript — a distributed systems project that decomposes large requests into dependency-aware task DAGs executed across multi-machine AI agents with retries, timeout handling, and fault-tolerant multi-threaded state management. This project demanded the same systems-level thinking — identifying performance bottlenecks, designing for reliability, and iterating on architecture — that I would bring to AWS engineering teams working on cloud computing infrastructure, distributed services, or AI/ML systems.

I have always been passionate about customer obsession and the principle of inventing and simplifying — building systems that are both technically rigorous and genuinely useful to the people who rely on them — and I envision myself contributing my talent to the furthering of AWS's mission to be Earth's Most Customer-Centric Company. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan